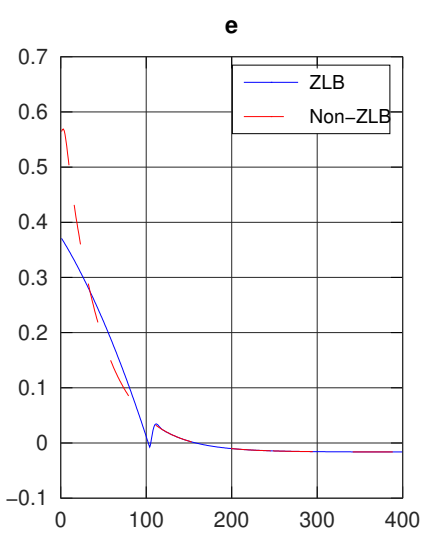
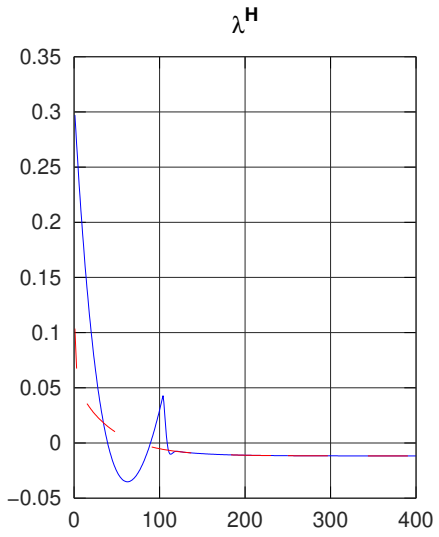
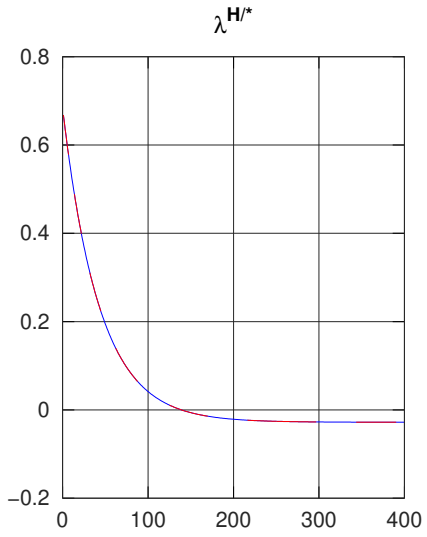
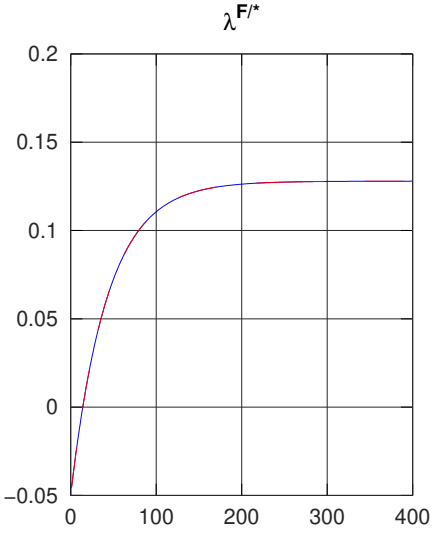
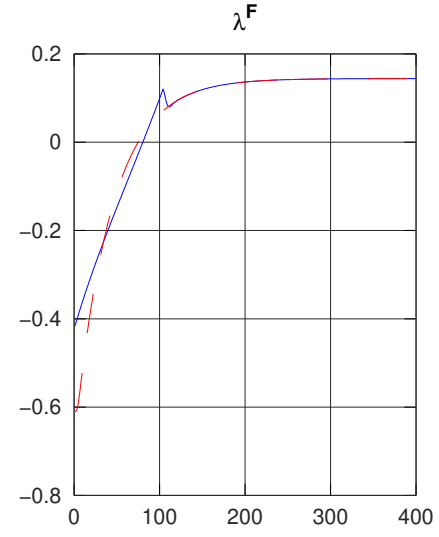
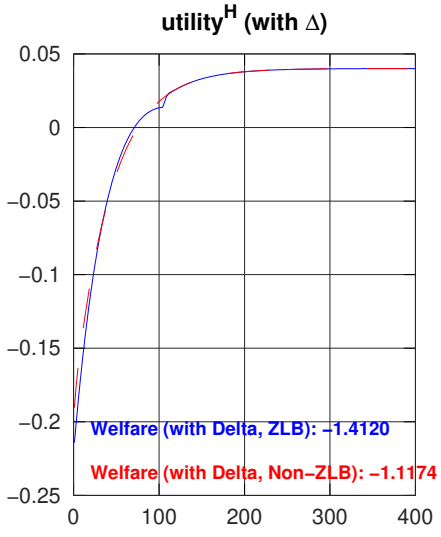

e
ZLB
Non−ZLB
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
−0.1
0
100
200
300
400

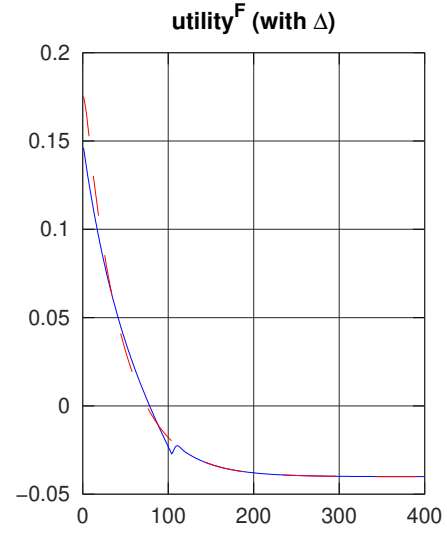

λ^H
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

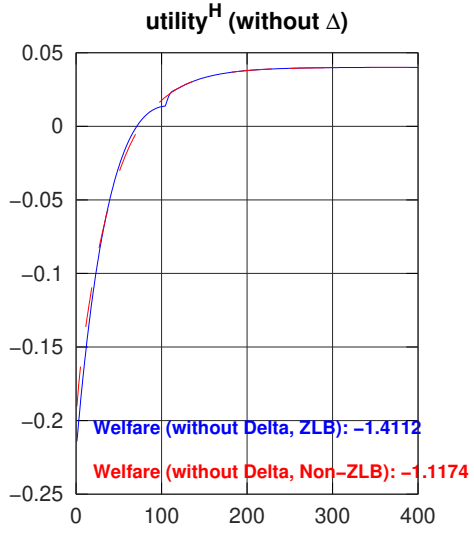

λ^H/*
0.8
0.6
0.4
0.2
0
−0.2
0
100
200
300
400

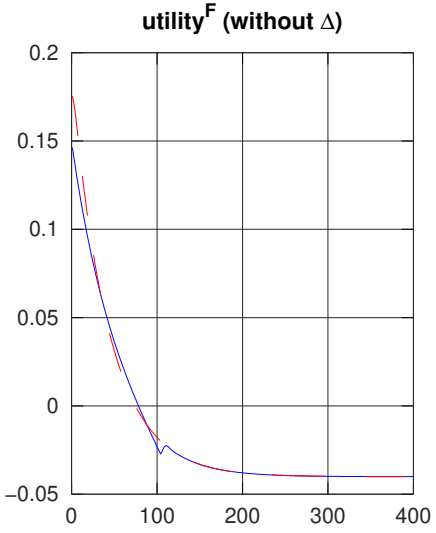

λ^F/*
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

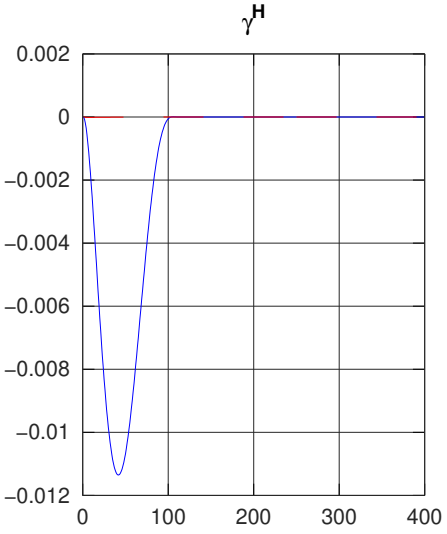

λ^F
0.2
0
−0.2
−0.4
−0.6
−0.8
0
100
200
300
400

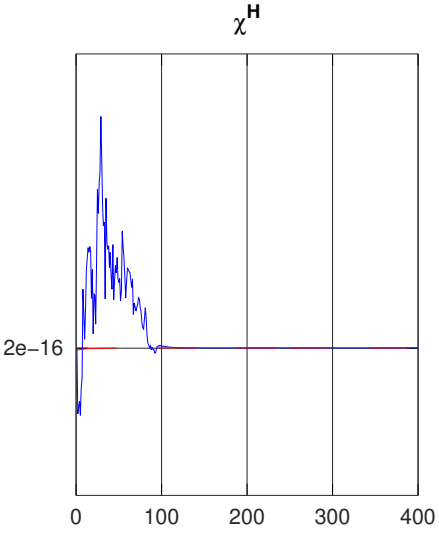

utility^H (with Δ)
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
0
100
200
300
400
Welfare (with Delta, ZLB): −1.4120
Welfare (with Delta, Non−ZLB): −1.1174

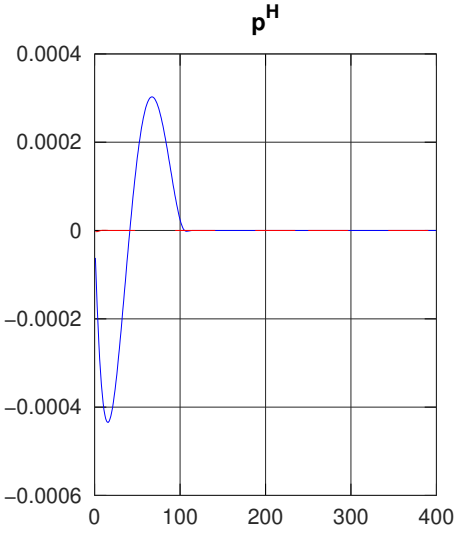

utility^F (with Δ)
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

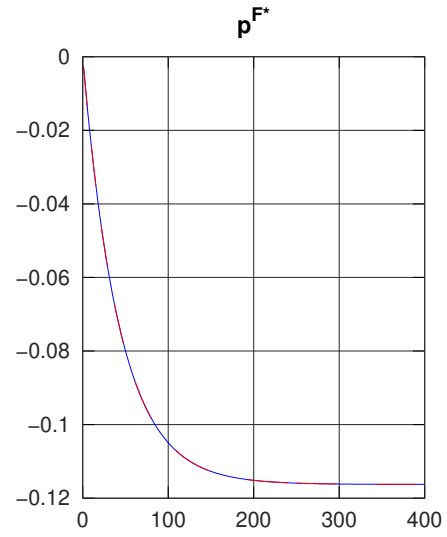

utility^H (without Δ)
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
Welfare (without Delta, ZLB): −1.4112
Welfare (without Delta, Non−ZLB): −1.1174
0
100
200
300
400

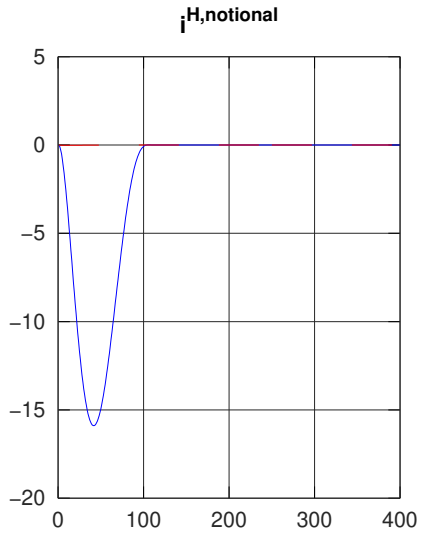

utility^F (without Δ)
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

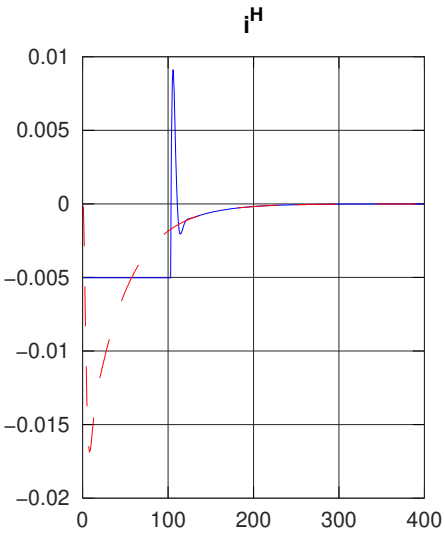

γ^H
0.002
0
−0.002
−0.004
−0.006
−0.008
−0.01
−0.012
0
100
200
300
400

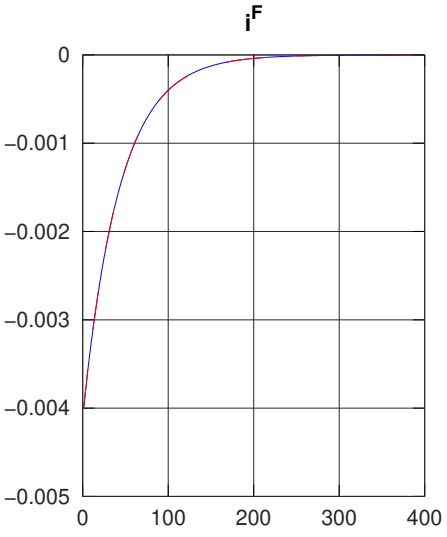

χ^H
2e−16
0
100
200
300
400

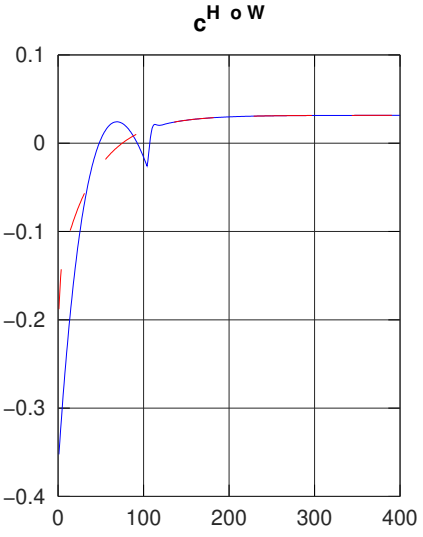

p^H
0.0004
0.0002
0
−0.0002
−0.0004
−0.0006
0
100
200
300
400

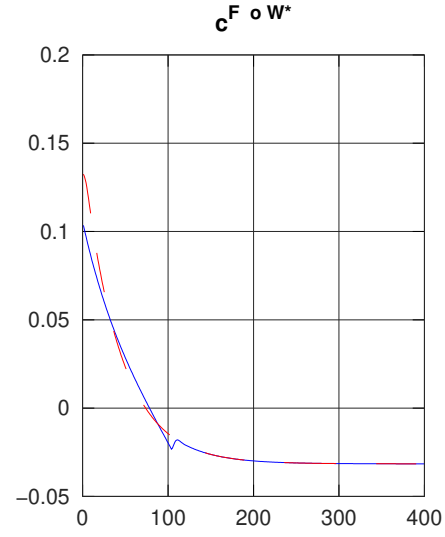

p^{F*}
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

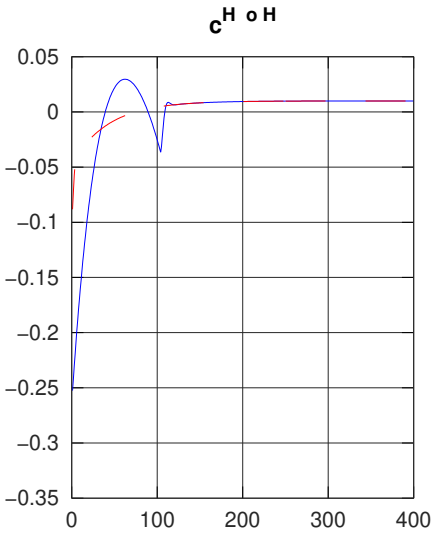

$i^{H,notional}$
5
0
−5
−10
−15
−20
0
100
200
300
400

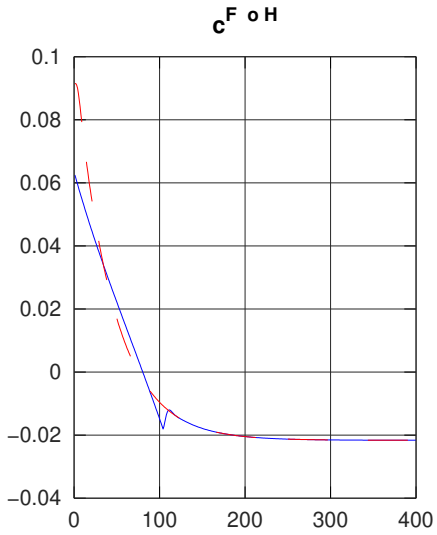

i^{H}
0.01
0.005
0
−0.005
−0.01
−0.015
−0.02
0
100
200
300
400

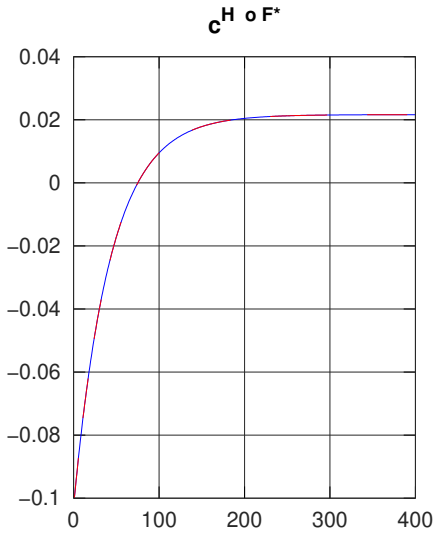

i^{F}
0
−0.001
−0.002
−0.003
−0.004
−0.005
0
100
200
300
400

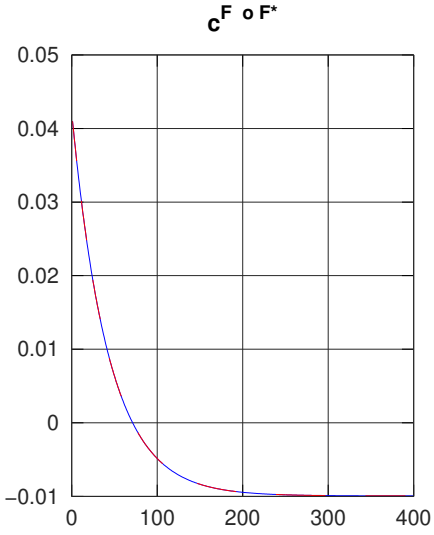

$c^{H\ o\ W}$
0.1
0
−0.1
−0.2
−0.3
−0.4
0
100
200
300
400

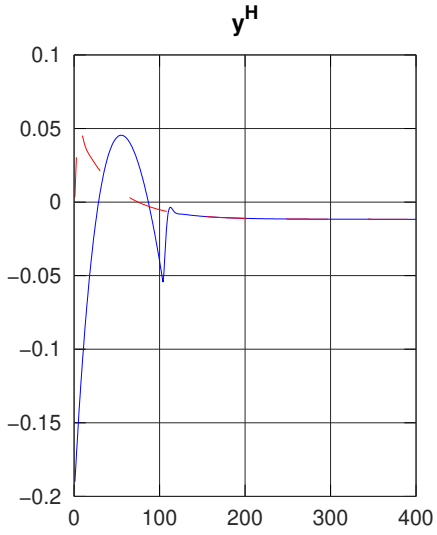

$c^{F\ o\ W*}$
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

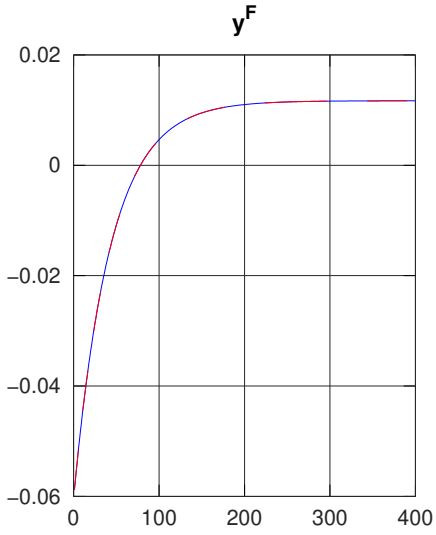

$c^{H\,o\,H}$
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
−0.3
−0.35
0
100
200
300
400

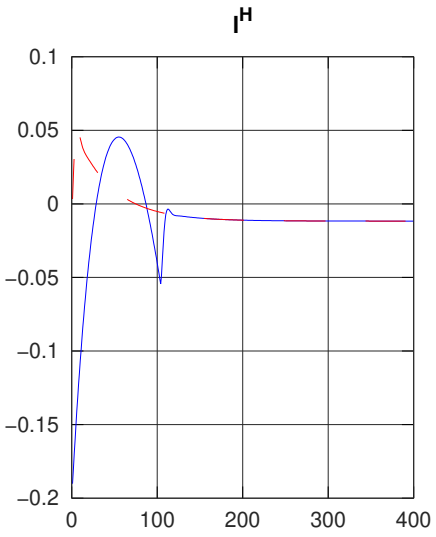

$c^{F\,o\,H}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
−0.04
0
100
200
300
400

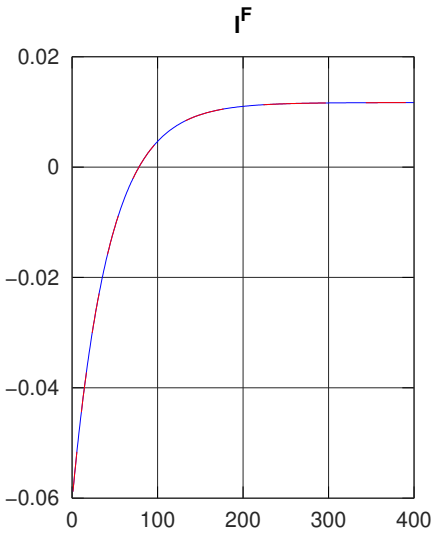

$c^{H\,o\,F^*}$
0.04
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.1
0
100
200
300
400

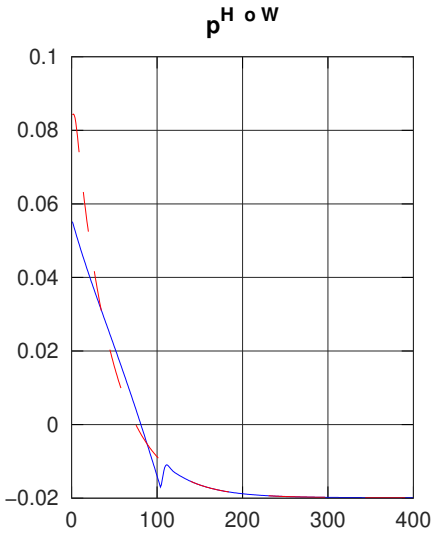

$c^{F\,o\,F^*}$
0.05
0.04
0.03
0.02
0.01
0
−0.01
0
100
200
300
400

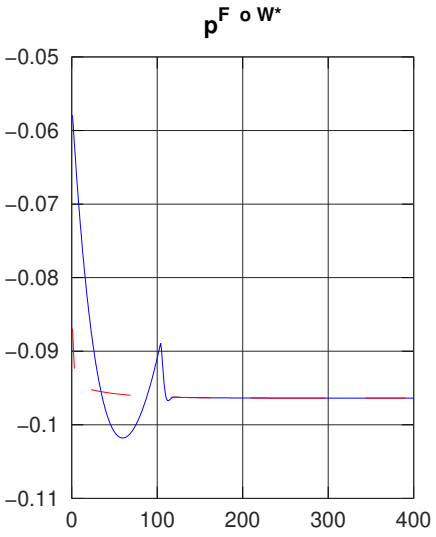

y^{H}
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400

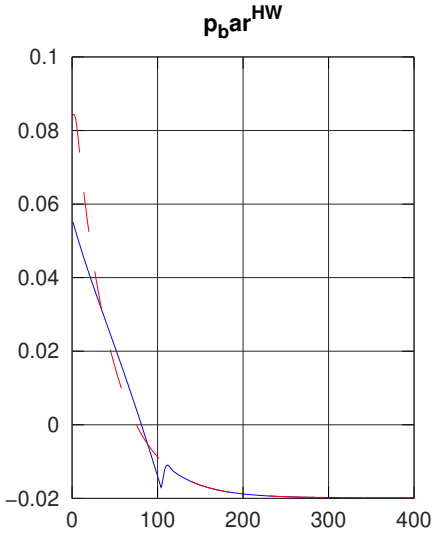

y^{F}
0.02
0
−0.02
−0.04
−0.06
0
100
200
300
400

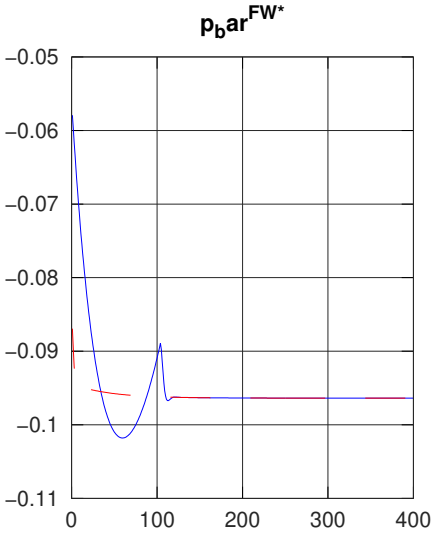

l^H
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400

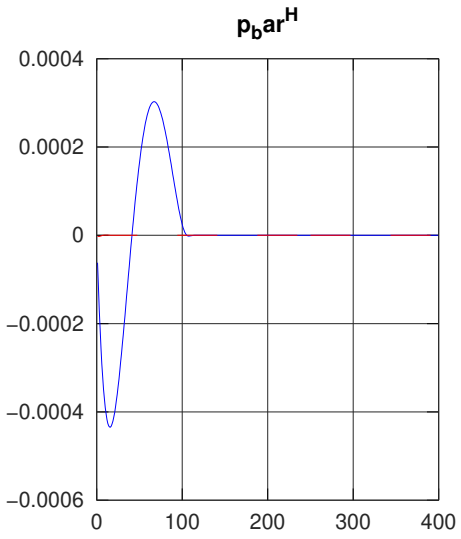

l^F
0.02
0
−0.02
−0.04
−0.06
0
100
200
300
400

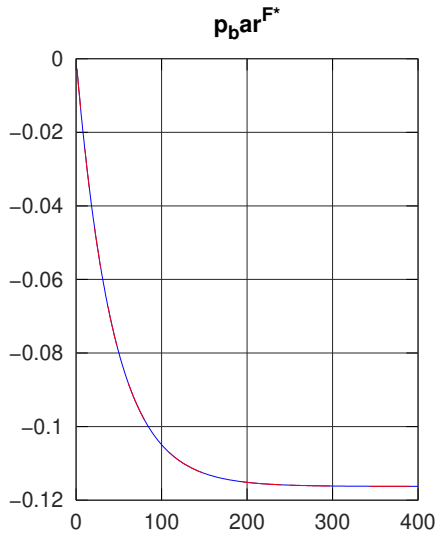

$p^{H\ o\ W}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

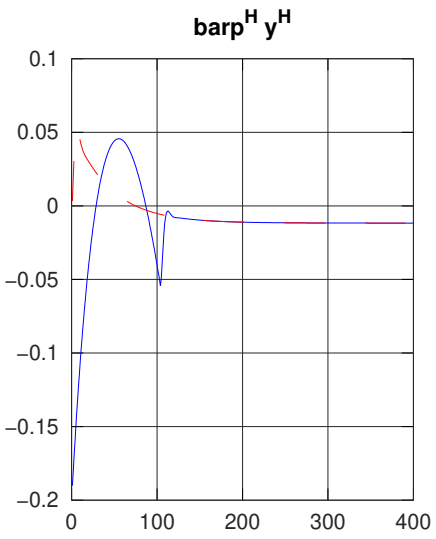

$p^{F\ o\ W^*}$
−0.05
−0.06
−0.07
−0.08
−0.09
−0.1
−0.11
0
100
200
300
400

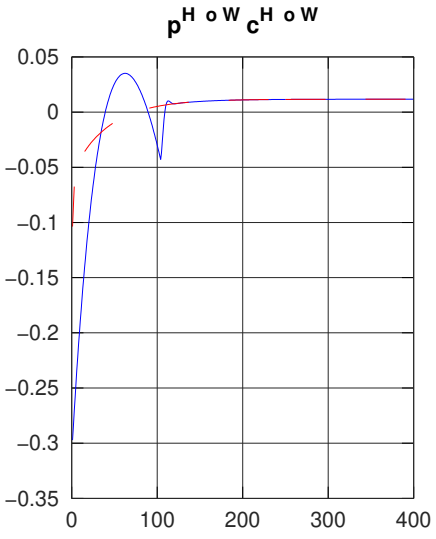

p_bar^{HW}
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

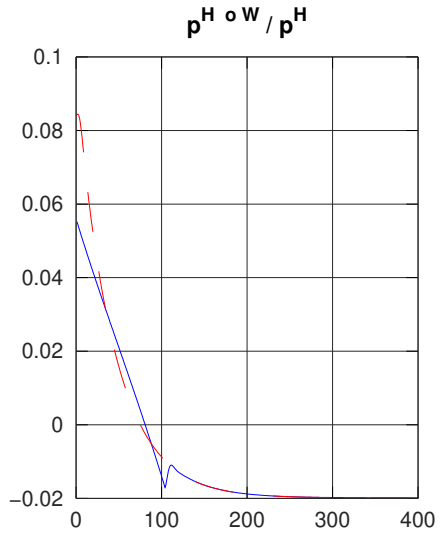

$p_bar^{FW^*}$
−0.05
−0.06
−0.07
−0.08
−0.09
−0.1
−0.11
0
100
200
300
400

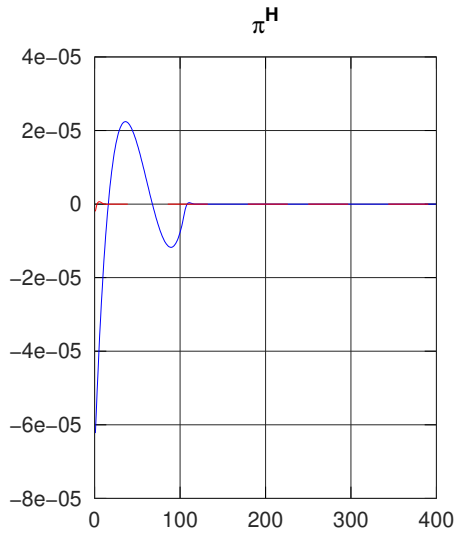

p_{bar}^{H}
0.0004
0.0002
0
−0.0002
−0.0004
−0.0006
0
100
200
300
400

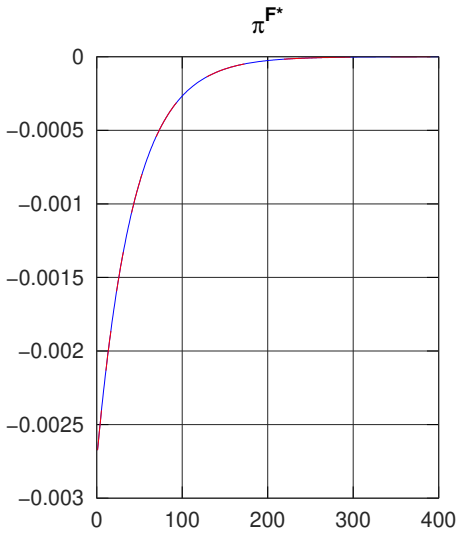

p_{bar}^{F*}
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

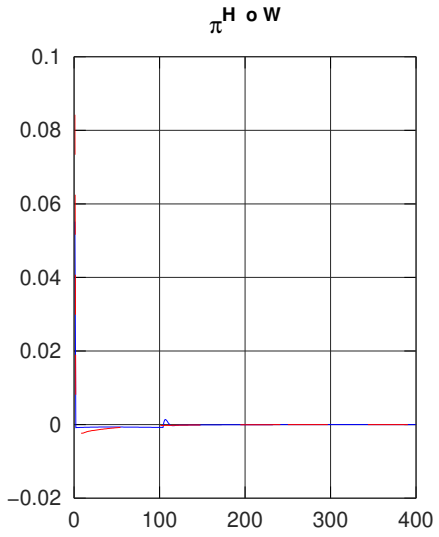

$barp^{H}\ y^{H}$
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400

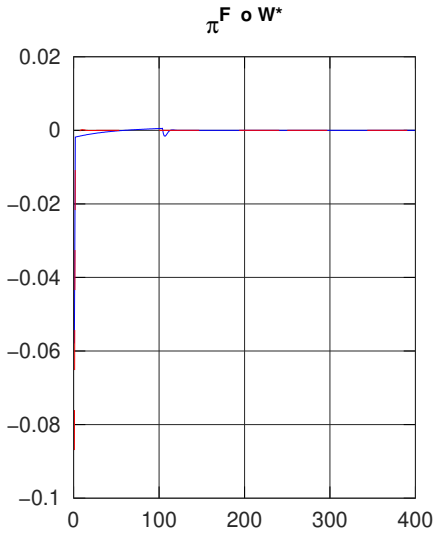

$p^{H\ o\ W}\ c^{H\ o\ W}$
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
−0.3
−0.35
0
100
200
300
400

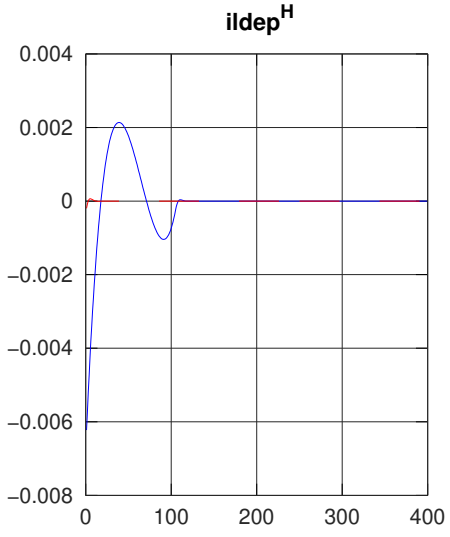

$p^{H\ o\ W}\ /\ p^{H}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

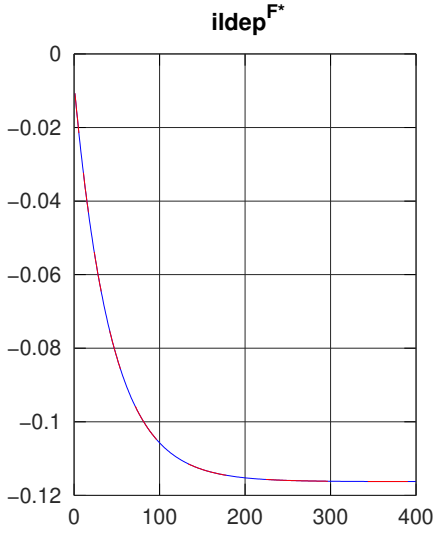

π^{H}
4e−05
2e−05
0
−2e−05
−4e−05
−6e−05
−8e−05
0
100
200
300
400

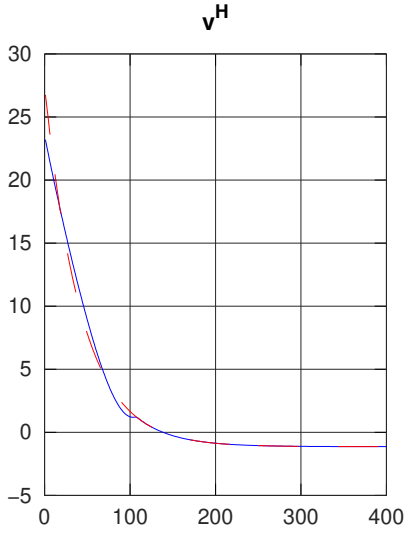

π^{F*}
0
−0.0005
−0.001
−0.0015
−0.002
−0.0025
−0.003
0
100
200
300
400

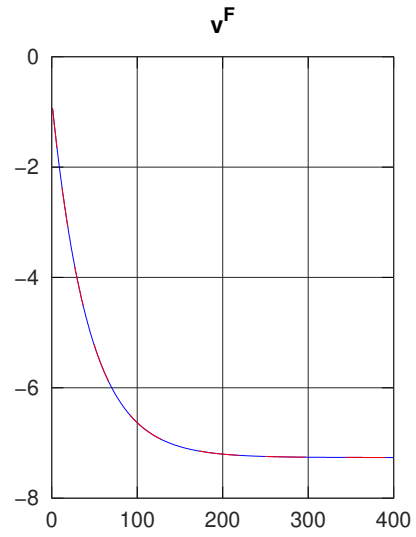

$\pi^{H\ o\ W}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

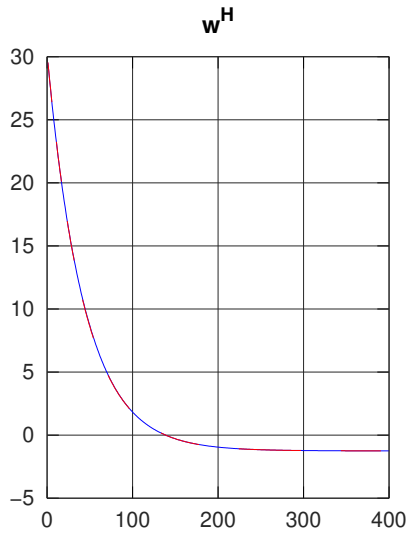

$\pi^{F\ o\ W*}$
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.1
0
100
200
300
400

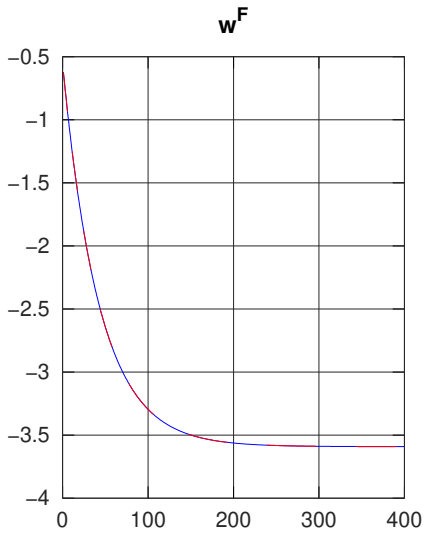

$ildep^{H}$
0.004
0.002
0
−0.002
−0.004
−0.006
−0.008
0
100
200
300
400

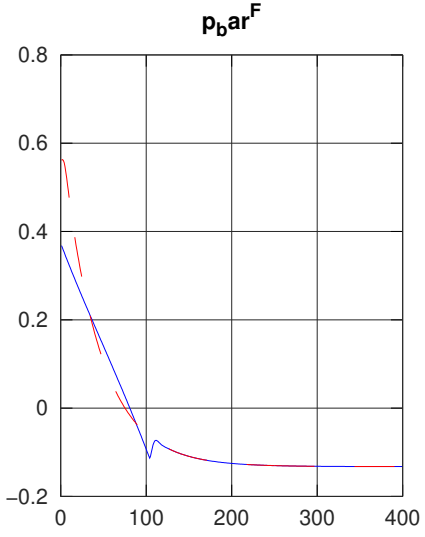

$ildep^{F*}$
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

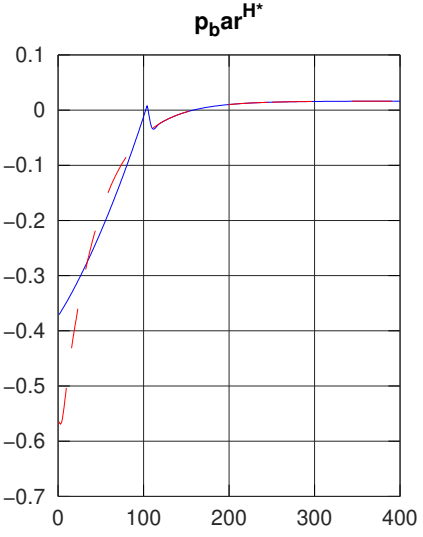

v^{H}
30
25
20
15
10
5
0
−5
0
100
200
300
400

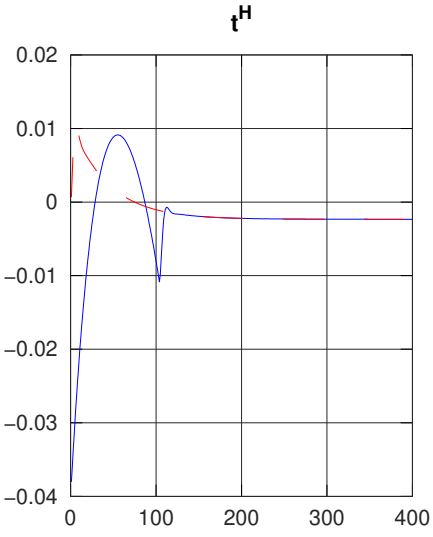

v^F
0
−2
−4
−6
−8
0
100
200
300
400

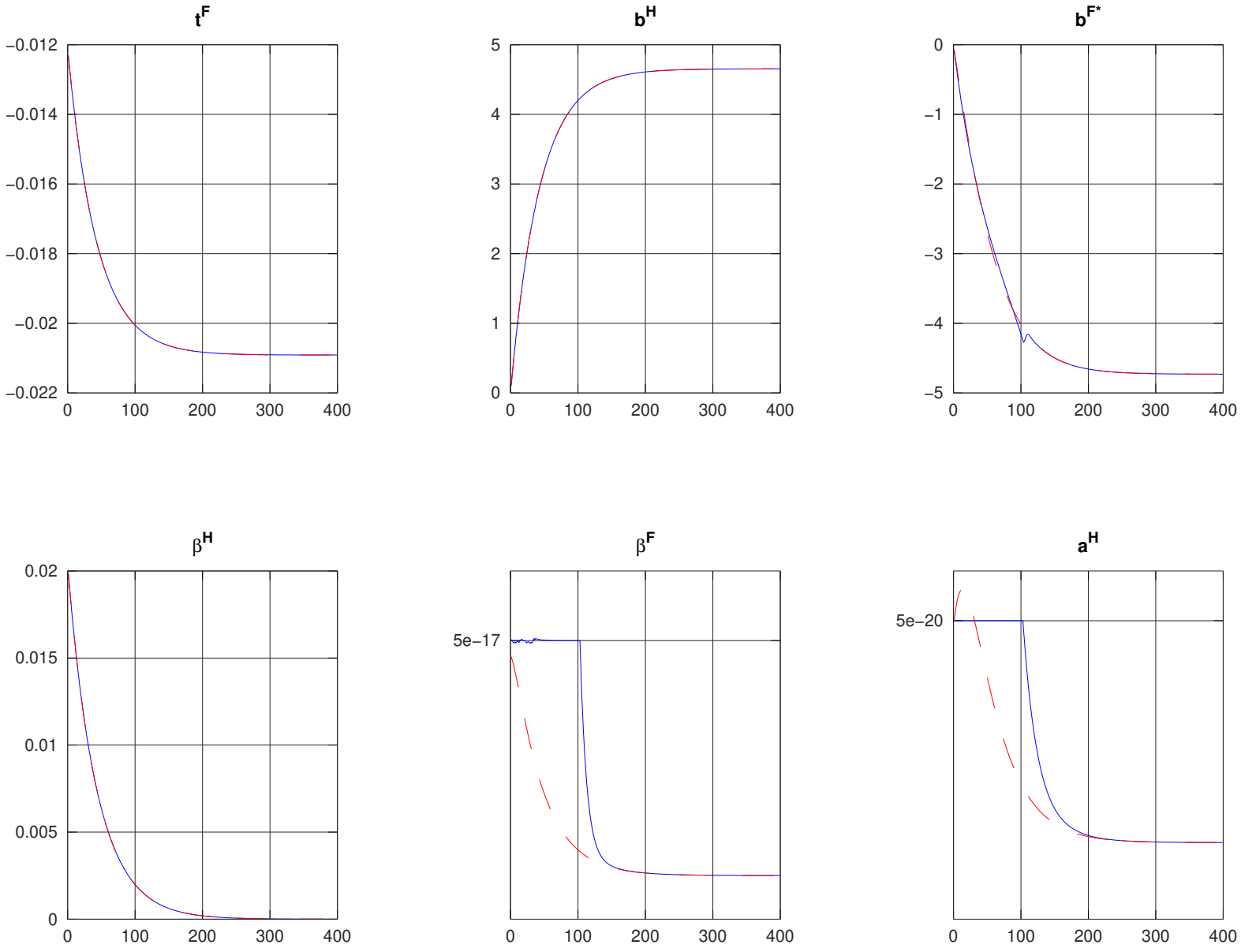

w^H
30
25
20
15
10
5
0
−5
0
100
200
300
400

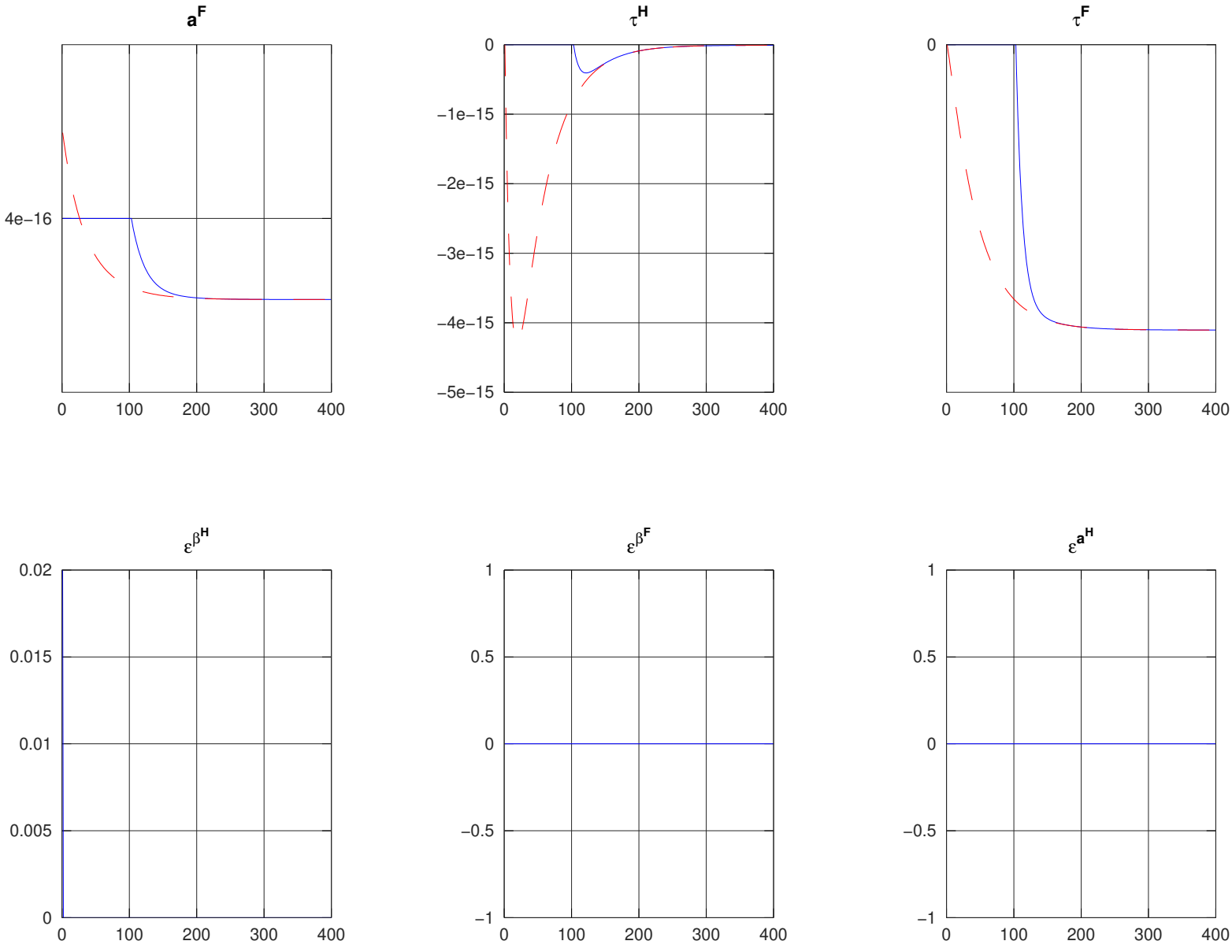

w^F
−0.5
−1
−1.5
−2
−2.5
−3
−3.5
−4
0
100
200
300
400

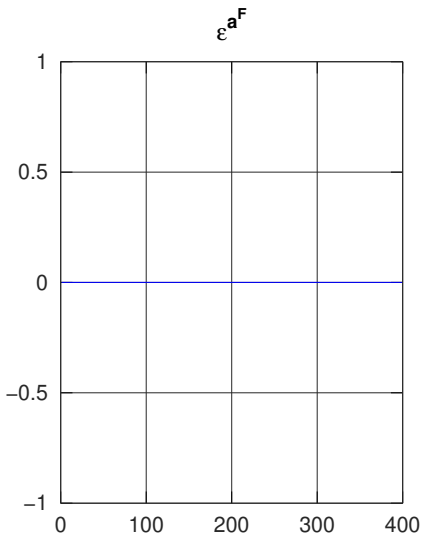

p_bar^F
0.8
0.6
0.4
0.2
0
−0.2
0
100
200
300
400

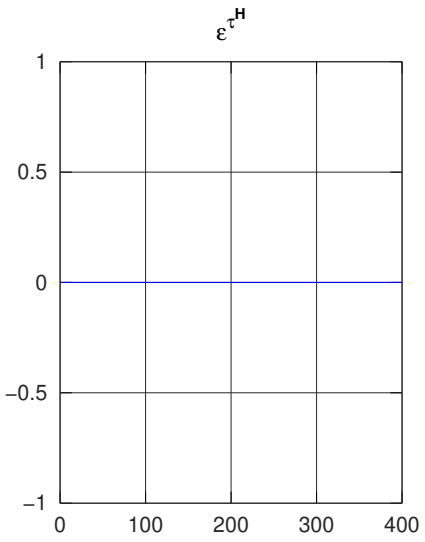

p_bar^{H*}
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
−0.7
0
100
200
300
400

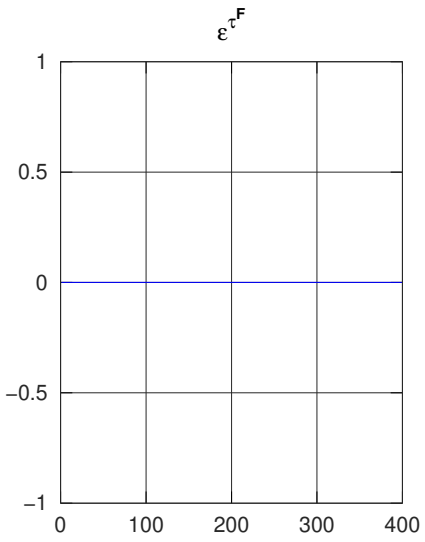

t^H
0.02
0.01
0
−0.01
−0.02
−0.03
−0.04
0
100
200
300
400

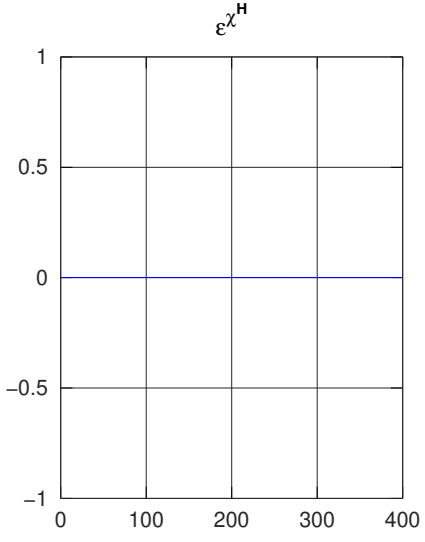

t^F
−0.012
−0.014
−0.016
−0.018
−0.02
−0.022
0
100
200
300
400
b^H
5
4
3
2
1
0
0
100
200
300
400
b^F*
0
−1
−2
−3
−4
−5
0
100
200
300
400
β^H
0.02
0.015
0.01
0.005
0
0
100
200
300
400
β^F
5e−17
0
100
200
300
400
a^H
5e−20
0
100
200
300
400

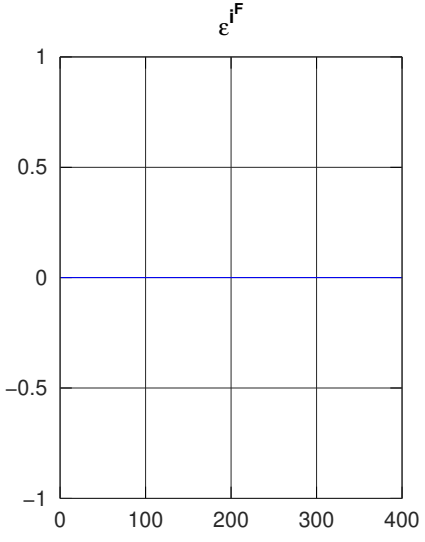

a^F
4e−16
0
100
200
300
400
τ^H
0
−1e−15
−2e−15
−3e−15
−4e−15
−5e−15
τ^F
ε^{β^H}
0.02
0.015
0.01
0.005
ε^{β^F}
1
0.5
−0.5
−1
ε^{a^H}

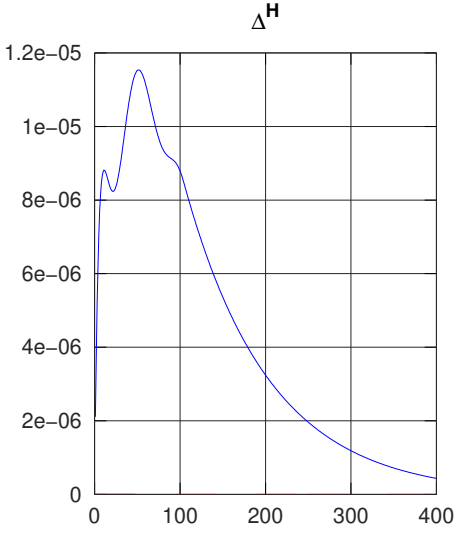

ε^{a^F}
1
0.5
0
−0.5
−1
0
100
200
300
400

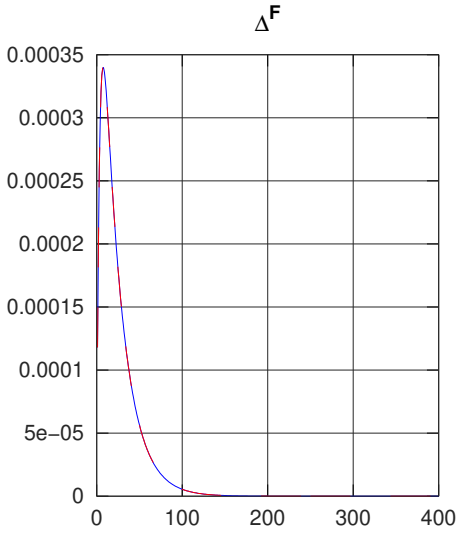

ε^{τ^H}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{τ^F}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{χ^H}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{i^F}
1
0.5
0
−0.5
−1
0
100
200
300
400


Δ^H
1.2e−05
1e−05
8e−06
6e−06
4e−06
2e−06
0
0
100
200
300
400


Δ^F
0.00035
0.0003
0.00025
0.0002
0.00015
0.0001
5e−05
0
0
100
200
300
400